

| EM for GMM |

"E"

$$\forall i: p(z_i | x_i, \theta^0) ; \forall i, \forall k p(z_i=k | x_i, \theta^0) = q_i(z_i=k)$$

"M"

$$\arg \max_{\theta} p(\{x_i\}_i, \underbrace{\{z_i\}_i}_{q(z)} | \theta)$$

	$\forall i$ z_i	θ
EM	$q_i(z_i)$	$\theta^0 \dots \theta^T$

$$\sum_{k \in \mathbb{R}^3} \pi_k = 1$$

$$\theta = \{\pi, \mu_1, \mu_2, \mu_3, \sigma_1, \sigma_2, \sigma_3\}$$

$\forall i$	$q_i(z_i)$	$q(\theta) = q_{\pi}(\pi) \prod_k q_k(\mu_k, \sigma_k)$
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$$q(\{z_i\}_i) = \prod_i q_i(z_i)$$

e.g. $q_{\pi} = \text{Dirichler}(\alpha)$

variational
parameters

$$(1000, 10, 10)$$

$$(100, 1, 1)$$

VAE

$$z_i \sim p_{\text{prior}}() \leftarrow \forall i, z_i \sim \mathcal{N}(0, I_d)$$

$$\forall i: x_i \sim \mathcal{N}(\text{dec}_\theta(z_i), I_d)$$

MAP:

$$\arg \max_{\mathbf{z}} \log p(\mathbf{z} | X) = \arg \max \underbrace{\log p(X | \mathbf{z})}_{\log \text{ likelihood}} + \underbrace{\log p(\mathbf{z})}_{\log \text{ prior}} - \cancel{\log p(X)}$$

$$= \arg \max_i \sum_i \|x_i - \text{dec}(z_i)\|^2 - \lambda \cdot \|\mathbf{z}\|^2$$

VBI: $\forall i, q_i(z_i) = \mathcal{N}(\mu_i, \underline{\sigma_i} I_d)(z_i)$

$\{\mu_i, \sigma_i\}_i$ are the var. parameters.

Loss: $\min KL(q(z) \parallel p(z | X)) = E_q \left[\log \frac{q(z)}{p(z | X)} \right]$

$$= \sum_i E_{z_i \sim q_i(z_i)} \left[\frac{1}{2\sigma_i^2} \|x_i - \text{dec}(z_i)\|^2 \right] + \sum_i KL(q(z_i) \parallel p(z))$$

$$\mathcal{N}(\mu_i, \sigma_i^2)$$

AMORTIZATION:

replace $\{\mu_i, \sigma_i\}_i$

$$KL(\mathcal{N}_1, \mathcal{N}_2)$$

$$\mathcal{N}(\mu, \sigma, Id) \text{ prior } \mathcal{N}(0, Id)$$

$$\approx K + \underbrace{\|\mu_i - 0\|^2 - \log \sigma_i + \sigma_i}$$

by enc_φ s.t. $(\mu_i, \sigma_i) = \text{enc}_\varphi(x_i)$

$$\sum_i \mathbb{E}_{z_i \sim \mathcal{N}(\text{enc}_\varphi(x_i))} \left[\|x_i - \text{dec}(z_i)\|^2 \right] + \lambda \cdot KL(\mathcal{N}(\text{enc}_\varphi(x_i)) \parallel \mathcal{N}(0, Id))$$

sample
"minibatches"
SGD

sample

approximates with 1 sample

$$\mathbb{E}_{z_i \sim \mathcal{N}(\dots)} [\boxtimes]$$

$\Rightarrow \boxtimes$ with $z_i \sim \mathcal{N}(\dots)$

doubly stochastic optimization

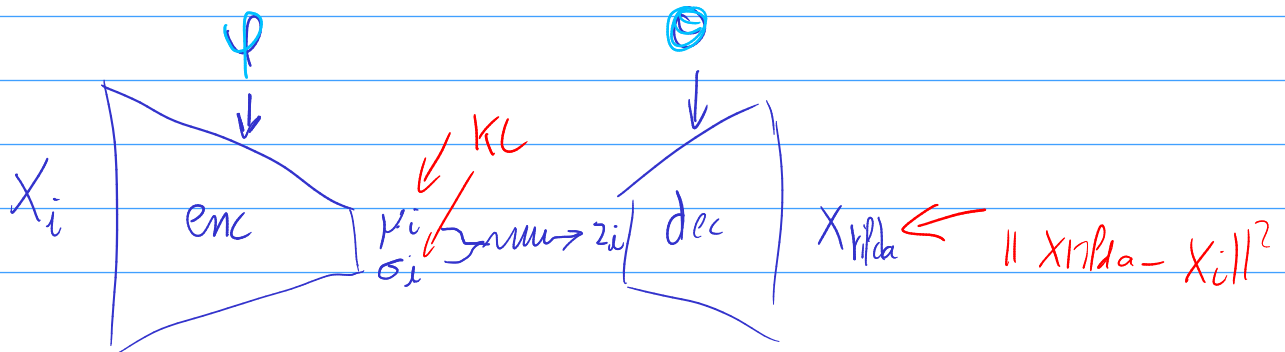
sample i

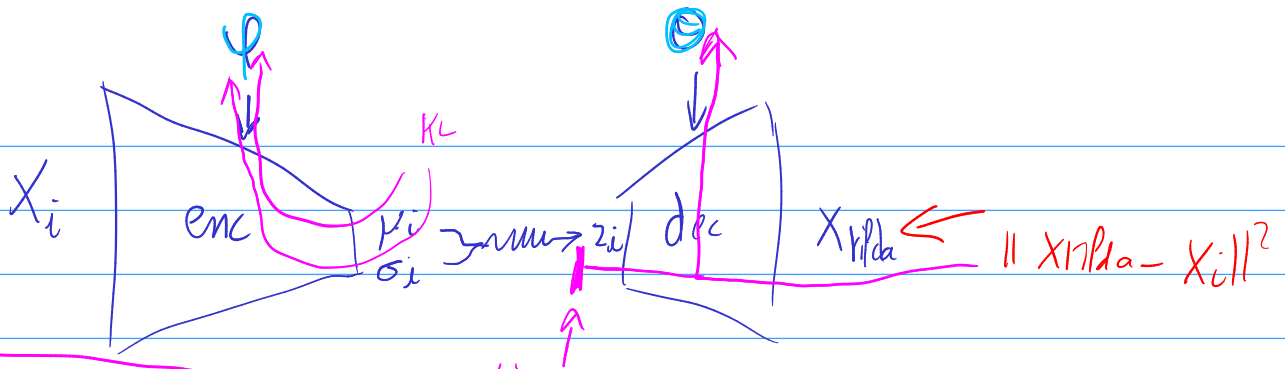
$$(\mu_i, \sigma_i) := \text{enc}_\varphi(x_i)$$

$$z_i := \text{randnormal}(\mu_i, \sigma_i)$$

$$x_{\text{filda}} = \text{dec}_\Theta(z_i)$$

$$\text{SGD step on } \|x_{\text{filda}} - x_i\|^2 + \lambda \underbrace{\left(\frac{\|\mu_i\|^2 + \sigma_i^2}{2} \log \sigma_i \right)}_{\text{KL}(\dots)}$$





$z_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$  gradient blocked

$\Leftrightarrow$

$$z_i = \sigma_i \epsilon_i + \mu_i, \text{ ; } \epsilon_i \sim \mathcal{N}(0, 1)$$

Reparameterization trick

